iii. Three phase AC system
   a) Three phase three wires
   b) Three phase four wires

1.3 Direct Transmission

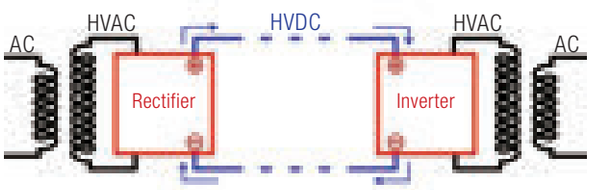


Fig. 1.1 Schematic diagram of DC transmission

Let us see the transmission of DC power from the AC power.

The single line diagram of high voltage direct current transmission is given in figure 1.1. Alternative current is generated and stepped up to high voltage through the sending end transformer. This high voltage alternating current is converted to the direct current by a mercury arc rectifier. The transmission of electric power is carried out at high DC voltage. At the receiving end, the DC voltage is converted into alternating current through the help of thyratrons. This alternating current is reduced to the low voltage through the receiving end transformer for distribution.

1.3.1 Advantages of DC transmission

i. Two conductors are sufficient for distribution of power supply
ii. There is no inductance and capacitance.
iii. No skin effect in DC transmission lines.
iv. The corona loss is lower as compared to AC system.
v. Voltage drop is very low.

1.3.2 Disadvantages of DC transmission

i. It is difficult to produce high voltage
ii. The voltage cannot be increased or decreased by using a transformer

1.4 Alternative transmission

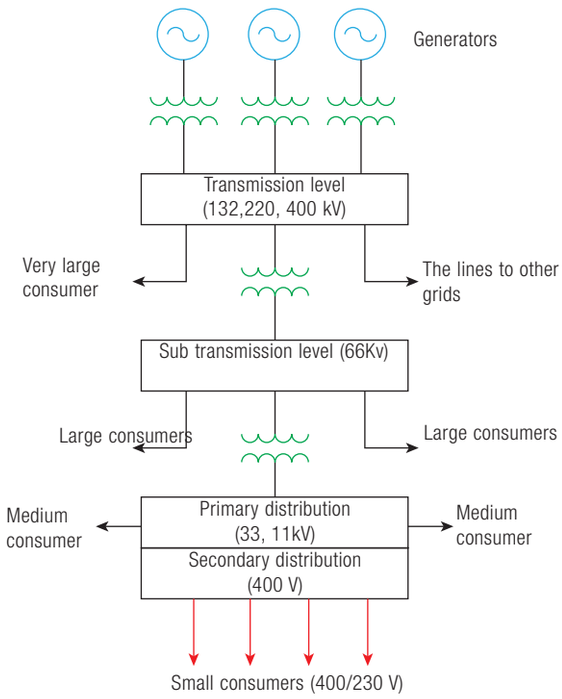


Fig. 1.2 AC power supply system

From the power generating stations, a large amount of AC supply is produced and is transmitted through transmission lines. Figure 1.2 shows the layout of a typical AC power supply system. The transmission is done caried by 3 phase 3 wire and 3 phase 4 wire systems. Other methods are also used for some special reasons.

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